

External validation of a two-timescale adaptive-margin model of background-conditioned physiological vulnerability

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Abstract

The TwoTimescaleResilience framework models *background-conditioned vulnerability*: chronic environmental pressure and retained chemical burden slowly narrow a species' adaptive margin A_t , weakening its restoring force $\lambda(A_t)$ and amplifying the burden of a later acute perturbation. Capacity parameters are derived offline from Add-my-Pet (AmP/DEB) records, pressure from EPA ECOTOX, and memory from a compound-retention recurrence. Here we confront the recovery/margin layer with *independent* external data at multiple organisational levels, using rank statistics and pre-registered effect signs throughout. (i) At the **population** level, the DEB maintenance rate constant k_M predicts demographic recovery (damping ratio) in the COMADRE matrix database beyond pace-of-life and a real dated phylogeny ($\rho=+0.22$, $p=0.011$, rank PGLS), and the DEB reproduction rate R_i predicts demographic compensation ($\rho=+0.77$). (ii) At the **individual** level, the modelled margin tracks Scope for Growth where tissue burden indexes exposure (Widdows 1995, $\rho=+0.41$), survival-in-air under emersion (ICES DOME, $\rho=+0.39 \rightarrow +0.45$ under body-size and condition control), and a controlled dose-response and additive mixture (Viarengo 1995). (iii) The erosion **dynamics** reproduce continued multi-month decline that a static map cannot (transplant proof-of-concept). The one-dimensional amplification scalar $g/\mathcal{F}$ is **null in every test**, as the margin-first reading predicts. Two well-powered **negative controls** bound the claim: the single-trait link from k_M to acute toxic sensitivity ($n=310$) is fully accounted for by body size, so the model's distinctive leverage is its across-axis *capacity weighting*, not k_M alone. Effects are rank-robust, magnitude-modest, and specification-sensitive—corroboration of the margin/recovery layer, honestly bounded.

1 Introduction

The framework and its mathematics are developed elsewhere (the companion paper; also Kooijman, 2010). In brief, for a target species it maps environmental pressure to burdens on four DEB physiological axes (assimilation, maintenance, growth, reproduction), aggregates these into a scalar *adaptive margin* A_t , and lets A_t set a slow restoring force $\lambda(A_t)$ whose two-timescale interplay with acute perturbation defines vulnerability. The capacity layer is parameterised offline from AmP (Marques et al., 2018), pressure from ECOTOX (Olker et al., 2022), and memory from a compound-retention recurrence.

A model assembled from independent ingredients is testable without circularity provided the *outcome* side never uses a model-derived quantity. We therefore confront the framework with external data at several organisational levels, asking a deliberately conservative question at each: does a model quantity predict an independently measured outcome, *beyond* the obvious confounds (pace-of-life, phylogeny, body size)? Three commitments keep the test honest: (a) **rank statistics** throughout (unit-invariant, robust to the monotone-but-nonlinear responses the model produces); (b) **pre-registered effect signs**; and (c) explicit reporting of nulls and bounds.

A central interpretive point—the *margin-first* reading—is that the validated product is the margin/recovery layer (its rate endpoints, state, function, and dynamics), *not* the one-dimensional amplification scalar $\mathcal{F} = \lambda(A_0)/\lambda(A_t)$, which by construction can carry only one physiological number per species and which we find predicts nothing externally.

2 Methods

Static (point) response. Every cross-sectional test uses the instantaneous map `compute_adaptive_margin_response` per-axis pressure x_i (a threshold-free, median-normalised relative tissue burden, $x_i = c_i/\text{median}(c_i)$) becomes a bounded impairment $E_i = x_i/(1 + x_i)$; impairments aggregate as $Q = \sum_i w_i E_i$ with AmP-derived weights w_i ; and the margin is $A_t = A_0(1 - Q)$. This exercises the response-curve shape, the mode-of-action (MoA) axis routing, and the AmP capacity weighting, but not the time dynamics.

Dynamic response. The transplant test integrates the slow erosion state, which relaxes at rate $\lambda \in [k_M, k_M g]$ with time-constant $1/\lambda$; for *M. edulis* $1/\lambda \approx$ months, *unfitted* (it falls out of the maintenance rate $k_M = [p_M]/[E_G]$).

MoA routing. Contaminants route to axes by physiological mode of action: hydrocarbons/PAH $\rightarrow$ assimilation; metals $\rightarrow$ maintenance (a deliberate confound-control for field tissue burden; see §4); organotin/organochlorine $\rightarrow$ reproduction; dibutyltin $\rightarrow$ growth.

Controls and statistics. Recovery endpoints are controlled for pace-of-life (generation time) and phylogeny (taxonomic-Order proxy and a dated-tree PGLS with Pagel’s λ by maximum likelihood; Pagel, 1999); individual-level energetics are controlled for body size and condition. All correlations are Spearman rank coefficients; $*p < 0.05$, $**p < 0.01$.

Data anchors. COMADRE (Salguero-Gómez et al., 2016); GlobTherm (Bennett et al., 2018); Scope for Growth (Widdows et al., 1995, 2002; Albentosa et al., 2012); survival-in-air (ICES DOME OSPAR CEMP, ICES, 2024) and controlled stress-on-stress (Viarengo et al., 1995); transplant and single-contaminant time-courses (Veldhuizen-Tsoerkan et al., 1991b,a); toxicokinetics (Rubach et al., 2010) and DEBtox/GUTS calibrations (Nyman et al., 2012; Ashauer et al., 2016; Baas and Kooijman, 2015; Jager, 2020; Jager et al., 2011).

3 Results

3.1 Recovery rate endpoints — COMADRE (population demography)

The DEB maintenance rate $\lambda_{\min} = k_M$ predicts the COMADRE log damping ratio (recovery rate) across $n=197$ matched species: raw $\rho = +0.41^{**}$, partial on generation time $+0.26^{**}$, and within-Order $+0.19^*$ — i.e. beyond pace and coarse phylogeny. A *dated* TimeTree (Kumar et al., 2017) PGLS over 182 species resolves the phylogeny question: Pagel’s $\lambda \approx 0$ (the trait carries little phylogenetic signal), and in rank form $k_M \rightarrow$ recovery **survives** the generation-time control under the dated covariance ($\beta^*=0.221$, $p=0.011$), while the log-linear form nulls ($p=0.96$). The effect is thus genuinely rank/monotone, robust to pace and real phylogeny simultaneously. The per-axis reproduction rate R_i predicts demographic *compensation* $\rho = +0.77^{**}$ (beyond pace and size) — the strongest single result. The amplification scalar g is null at every level (-0.05 to -0.13).

3.2 Capacity coherence and its bound — GlobTherm

In a pre-registered probe ($n=664$ AmP $\leftrightarrow$ GlobTherm species), the AmP capacity axis correlates with independent thermal-tolerance data ($|\rho|$ up to 0.45) — the offline AmP $\rightarrow$ capacity extraction carries real biological structure. But higher $k_M/\lambda_{\max}$ implies *narrower* thermal breadth (-0.45 all-taxa), **refuting** a general-resilience reading: recovery capacity is specific to demographic recovery, not a universal resilience currency.

3.3 The margin state — Scope for Growth

At the margin’s own organisational level (no scale bridge), the modelled margin tracks measured SFG along contaminant gradients in a scale-dependent way: $\rho = +0.41^*$ where tissue burden indexes exposure (Widdows 1995, estuary), $+0.12$ at basin scale (Widdows 2002), and -0.11 where condition/food dominates (Albentosa 2012; the authors find SFG $\sim$ condition -0.62^{***}). Across all three, metals behave as a *positive* confound (tissue burden tracks condition, not exposure); the MoA routing keeps them off the toxic axis, so the routed margin beats naive equal-weight load.

3.4 The margin function — stress-on-stress

Survival-in-air under emersion is the closest *outcome* to the margin’s purpose. In the ICES DOME panel (*M. edulis*, 17 UK stations), survival tracks the modelled margin $\rho = +0.39 \rightarrow +0.45$ under body-length and condition control. Decisively, and unlike Albentosa, *controlling* the condition confound here *strengthens* the signal — genuine margin erosion, not a health-proxy artifact. A controlled-exposure cross-check (Viarengo et al., 1995) gives a monotone dose-response (potency Cu>PAH>PCB) and a binary mixture that is additive with no antagonism, bracketed by the model’s own concentration-addition / independent-action rules.

3.5 The margin dynamics — transplant proof-of-concept

The only test that runs the erosion dynamics (Veldhuizen-Tsoerkan et al., 1991b): cadmium plateaus by 2.5 months yet survival keeps dropping to 5 months. A static map predicts no further erosion; the dynamic state rises $\sim 33\%$ on the model’s *unfitted* months-scale $1/\lambda$, matching the continued decline ($n=4$, proof-of-concept). A single-contaminant firm-up (Veldhuizen-Tsoerkan et al., 1991a) removes the PCB confound: cadmium alone erodes survival progressively, modelled margin tracking it $\rho = +0.90$.

3.6 The consistent null — the amplification scalar

Across *every* anchor the one-dimensional amplification scalar $g/\mathcal{F}$ predicts nothing. This is the margin-first prediction confirmed: external support is for the margin/recovery layer, not for a scalar amplification readout.

3.7 Bounding negative controls — single-trait $k_M \rightarrow$ toxicity is size-confounded

Two direct, out-of-sample tests of “does maintenance pace predict toxic response?” both reduce to a body-size confound. (i) *State axis*: species-level acute LC50/EC50 for four AChE inhibitors from EPA ECOTOX, matched to AmP ($n=310$; replicating Baas and Kooijman, 2015). Higher $k_M/[p_M] \rightarrow$ more sensitive ($\rho \approx -0.27$, all four chemicals) *raw*, but the partial controlling body size is ≈ -0.03 — a robust null at $n=310$. (ii) *Rate axis*: against measured chlorpyrifos elimination rate constants (Rubach et al., 2010), k_M is weak/wrong-signed and non-significant; and elimination is a

Table 1: Honest scorecard. Effects are Spearman ρ ; * $p < 0.05$, ** $p < 0.01$.

Anchor	Level	Result	
COMADRE (k_M , R_i)	population	$k_M \rightarrow \text{recovery} +0.19 \rightarrow +0.22^*$ (rank, dated PGLS); $R_i \rightarrow \text{comp.} +0.77^{**}$	c
GlobTherm ($n=664$)	physiology	capacity coherent ($ \rho \leq 0.45$), recovery-specific	b
Scope for Growth ($\times 3$)	energetics	$+0.41 \rightarrow +0.12 \rightarrow -0.11$ (scale-dependent)	c
Stress-on-stress (DOME)	energetics	$+0.39 \rightarrow +0.45$ (confound-controlled)	s
Viarengo 1995	controlled dose	monotone dose-response; additive mixture	c
Transplant (Veldhuizen)	dynamics, over time	continued erosion static map cannot; Cd-alone $+0.90$	p
$k_M \rightarrow \text{toxicity}$ ($n=310$)	cross-species	raw -0.27 ; nulls under body size	b
amplification $g/\mathcal{F}$	—	null everywhere	n

toxicokinetic, not the thesis-relevant toxicodynamic recovery rate (Nyman et al., 2012). The reading: the model’s distinctive leverage is its across-axis *capacity weighting*, not k_M as a scalar predictor.

4 Discussion

External evidence supports the **margin/recovery layer** at four organisational levels — its rate endpoints, its state, its function, and a first sign of its dynamics (Figure 1) — while the amplification scalar is null throughout, exactly as the margin-first reframe predicts. Three honest qualifications define the result.

Rank-robust, magnitude-modest, specification-sensitive. Effects are $\rho \approx 0.2\text{--}0.45$ — corroboration, not strong prediction — and several live in the ranks (the headline k_M result survives a dated-tree PGLS in rank form but nulls in log-linear form). They should be reported as monotone tendencies.

Tissue burden $\neq$ exposure, and the metals lesson. The pressure proxy is a relative tissue burden; where burden tracks food/condition rather than exposure it fails (Albentosa). This is why metals route to maintenance in field data: it is a *confound-control*, not a mechanistic claim. Fitted DEBtox modes of action place metals on assimilation, but re-routing metals to assimilation *degrades every field anchor* (e.g. Widdows 1995 $+0.41 \rightarrow +0.30$; Widdows 2002 $+0.12 \rightarrow -0.14$). The two assignments therefore coexist by data regime: field measured burden uses the maintenance confound-control; controlled or exposure-derived burden may use the fitted assimilation mode.

The distinctive content is the capacity weighting, and it is carried as an assumption. Every individual-level test is single-species, so the AmP capacity weighting is held constant; and the single-trait $k_M \rightarrow \text{toxicity}$ link is body-size-confounded. The model’s distinctive across-axis capacity weighting therefore remains *untested*, awaiting across-species contaminant-gradient data that largely does not exist; it is stated as a model assumption, like the mixture rules.

Scope for water-quality application. The validated machinery (margin, saturating response, recovery dynamics) supports *relative*, mechanistically structured vulnerability statements — ranking sites and months under sustained pressure — rather than absolute risk thresholds or across-species predictions. That is the defensible use the present evidence licenses.

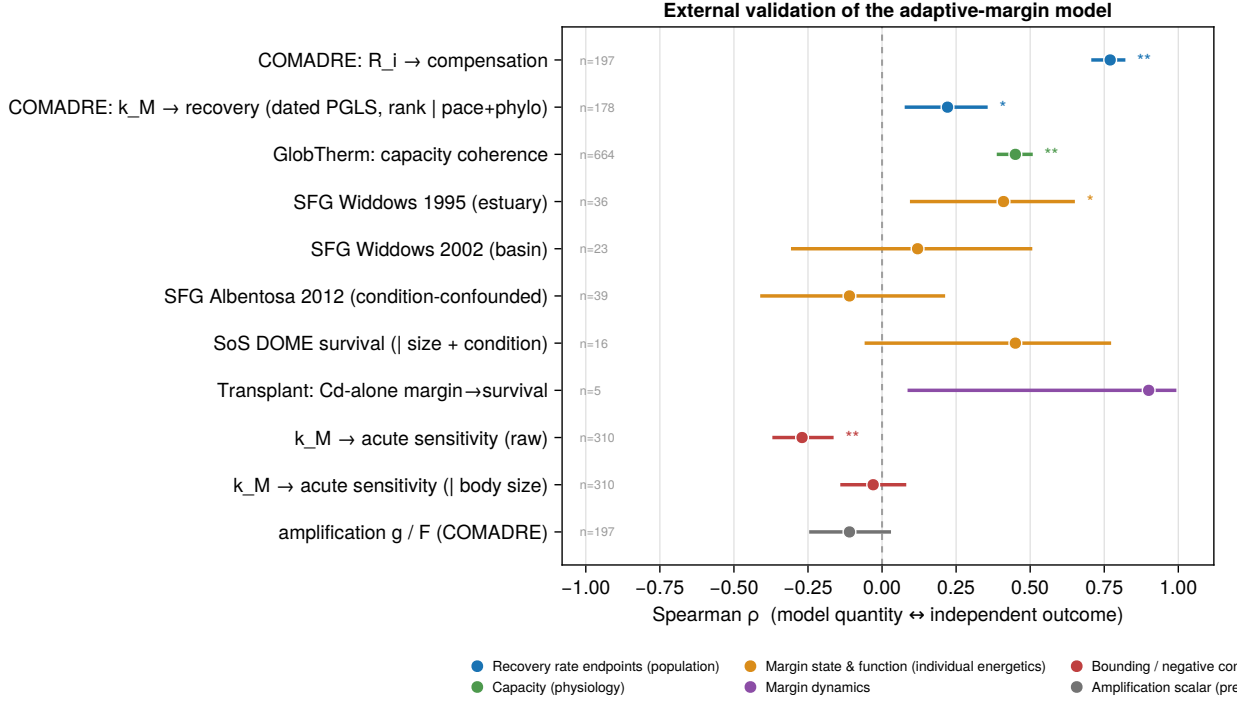


Figure 1: External-validation forest plot. Per-anchor Spearman ρ (point) with Fisher- z approximate 95% confidence intervals (bars), grouped by organisational level. Positive ρ indicates the model quantity tracks the independently measured outcome in the pre-registered direction. The positive cluster (recovery endpoints, margin state/function, dynamics) contrasts with the *bounding* negative controls — $k_M \rightarrow$ acute sensitivity is non-trivial raw but collapses under a body-size control ($n=310$) — and the amplification scalar $g/\mathcal{F}$, which is null as the margin-first reading predicts. CIs are illustrative (some anchors are small- n or figure-digitised); $*p < 0.05$, $**p < 0.01$.

Reproducibility and data availability

All anchors use openly available data (COMADRE, GlobTherm, ICES DOME OSPAR CEMP, EPA ECOTOX, AmP) or published tables; raw downloads are gitignored and derived tables, harnesses, and analysis scripts are in the repository (`examples/`, `scripts/`, `data/external/`). A consolidated account with per-anchor detail is `docs/notes/external_validation_synthesis.md`. Statistics are Spearman rank correlations with pre-registered signs; analyses run under Julia 1.12.6.

References

- M. Albentosa et al. Response of two mussel populations to contamination: scope for growth and biomarkers. *Science of the Total Environment*, 435–436:430–438, 2012.
- R. Ashauer et al. Modelling survival: exposure pattern, species sensitivity and uncertainty. *Scientific Reports*, 6:29178, 2016.
- J. Baas and S. A. L. M. Kooijman. Sensitivity of animals to chemical compounds links to metabolic rate. *Ecotoxicology*, 24:657–663, 2015.

- J. M. Bennett et al. GlobTherm, a global database on thermal tolerances for aquatic and terrestrial organisms. *Scientific Data*, 5:180022, 2018.
- ICES. Data and results for the 2024 OSPAR CEMP assessment. ICES, figshare 27211422 (CC BY 4.0), 2024.
- T. Jager. Revisiting simplified DEBtox models for analysing ecotoxicity data. *Ecological Modelling*, 416:108904, 2020.
- T. Jager, C. Albert, T. G. Preuss, and R. Ashauer. General unified threshold model of survival—a toxicokinetic-toxicodynamic framework for ecotoxicology. *Environmental Science & Technology*, 45(7):2529–2540, 2011.
- S. A. L. M. Kooijman. *Dynamic Energy Budget Theory for Metabolic Organisation*. Cambridge University Press, 3rd edition, 2010.
- S. Kumar et al. TimeTree: a resource for timelines, timetrees, and divergence times. www.timetree.org, 2017.
- G. M. Marques et al. The AmP project: Comparing species on the basis of dynamic energy budget parameters. *PLoS Computational Biology*, 14(5):e1006100, 2018.
- A.-M. Nyman, K. Schirmer, and R. Ashauer. Toxicokinetic-toxicodynamic modelling of survival of *Gammarus pulex* in multiple pulse exposures to propiconazole. *Ecotoxicology*, 21:1828–1840, 2012.
- J. H. Olker et al. The ECOTOXicology knowledgebase: A curated database of ecologically relevant toxicity tests. *Environmental Toxicology and Chemistry*, 41(6):1520–1539, 2022.
- M. Pagel. Inferring the historical patterns of biological evolution. *Nature*, 401:877–884, 1999.
- M. N. Rubach et al. Toxicokinetic variation in 15 freshwater arthropod species exposed to the insecticide chlorpyrifos. *Environmental Toxicology and Chemistry*, 29(10):2225–2234, 2010.
- R. Salguero-Gómez et al. COMADRE: a global data base of animal demography. *Journal of Animal Ecology*, 85(2):371–384, 2016.
- M. B. Veldhuizen-Tsoerkan et al. Effects of cadmium and PCB exposure on stress indices in the sea mussel *Mytilus edulis*. *Archives of Environmental Contamination and Toxicology*, 20:259–265, 1991a.
- M. B. Veldhuizen-Tsoerkan et al. A field study on stress indices in the sea mussel, *Mytilus edulis*: application of the "stress approach" in biomonitoring. *Archives of Environmental Contamination and Toxicology*, 21:497–504, 1991b.
- A. Viarengo et al. Stress on stress response: a simple monitoring tool in the assessment of a general stress syndrome in mussels. *Marine Environmental Research*, 39:245–248, 1995.
- J. Widdows et al. Measurement of stress effects (scope for growth) and contaminant levels in mussels (*Mytilus edulis*) along the North Sea coast. *Marine Ecology Progress Series*, 127:131–148, 1995.
- J. Widdows et al. Measurement of stress effects (scope for growth) and contaminant levels in mussels along the Irish Sea coast. *Marine Environmental Research*, 53:327–356, 2002.